

1.13 SPIN ECHO FORMATION (180° RF REFOCUSING)

A 180° RF pulse reverses dephasing and refocuses spins, forming an echo at time TE.

WHAT IS A SPIN ECHO?

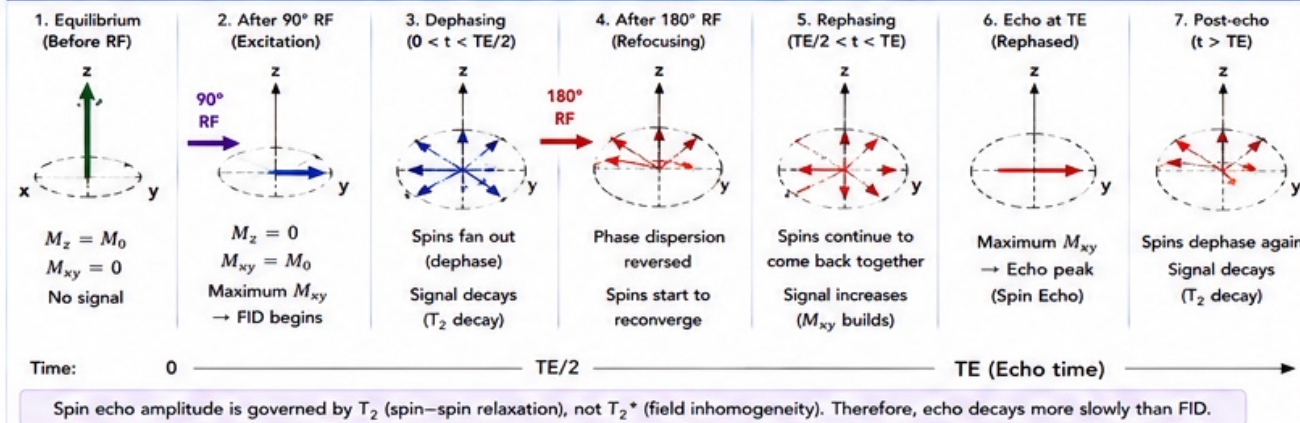
After a 90° RF pulse, spins fan out (dephase) due to microscopic field inhomogeneities. A 180° RF pulse flips the spins, reversing their phase dispersion.

Spins that were ahead are now behind, and those behind are now ahead. They rephase at time TE, producing a maximum signal — the spin echo.

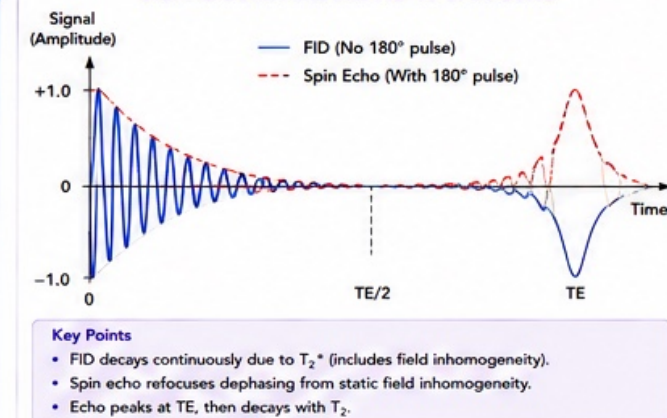
KEY IDEA

The 180° pulse acts like a mirror in time, undoing dephasing from T₂ effects (but not from T₂* field inhomogeneities).

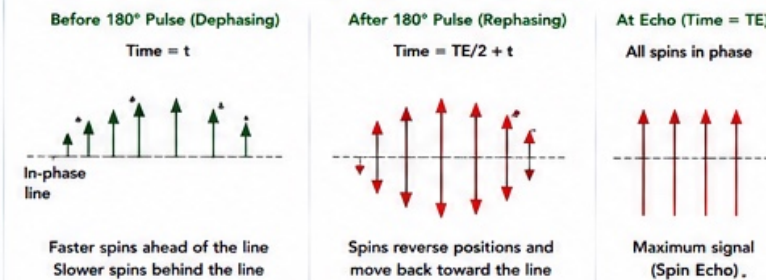
SPIN ECHO TIMELINE AND MAGNETIZATION BEHAVIOR



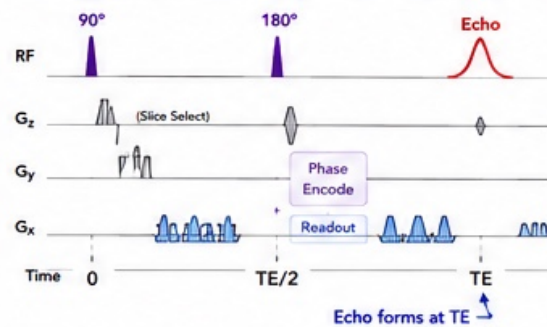
SIGNAL COMPARISON: FID vs SPIN ECHO



DEPHASING AND REPHASING VISUALIZED (IN-PHASE LINE)



PULSE SEQUENCE DIAGRAM (SPIN ECHO SEQUENCE)



TYPICAL T₂ VALUES AT 1.5T (APPROX.)

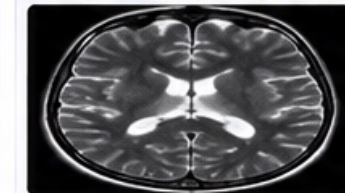
Tissue	T ₂ (ms)
Cerebrospinal Fluid (CSF)	80 – 120
Gray Matter	60 – 100
White Matter	60 – 90
Skeletal Muscle	40 – 60
Liver	30 – 50
Fat	20 – 40
Myocardium	35 – 55
Iron (hemosiderin)	< 10

Values vary with field strength, sequence, temperature, and vendor.

CLINICAL RELEVANCE (WHY USE SPIN ECHO?)

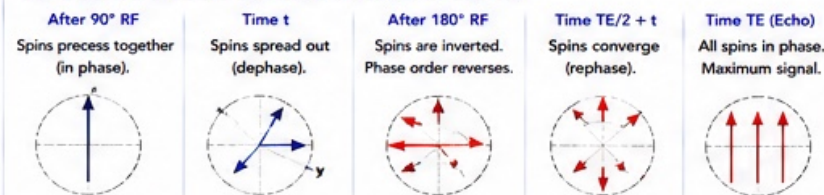
- Reduces effects of static field inhomogeneity → more consistent images.
- Basis for most SE, FSE, STIR, T₂-weighted and many quantitative sequences.
- Better for tissues near air-tissue or metal interfaces (less susceptibility artifact vs GRE).
- Essential for measuring true T₂ relaxation.

Example: Axial Brain (T₂-Weighted FSE)



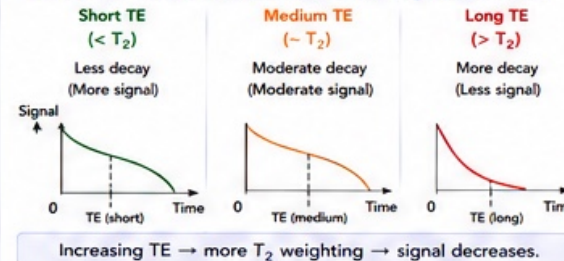
Long T₂ tissues (CSF) appear bright. Spin echo provides high contrast based on T₂ differences.

ECHO FORMATION EXPLAINED (PHASE PERSPECTIVE)



180° pulse reverses the dephasing process.

EFFECT OF TE ON SPIN ECHO SIGNAL (T₂ WEIGHTING)



COMMON MISCONCEPTION

"The 180° pulse eliminates all dephasing."

Reality: It refocuses dephasing caused by static field inhomogeneity (T₂* extra dephasing), diffusion, and chemical shift. True T₂ decay due to spin-spin interactions (microscopic fluctuations) is irreversible.

Spin echo removes T₂* effects but not true T₂ decay.

ENGINEERING INSIGHT

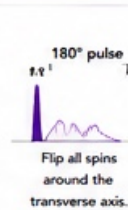
Echo quality depends on:

- Shim quality (B₀ uniformity)
- Gradient calibration & timing
- RF pulse accuracy (flip angle, slice profile)
- Coil integrity (preamp, cables, tuning)
- Eddy currents and gradient distortion



FIELD SERVICE TIP

- Check 180° flip angle calibration regularly.
- Verify gradient timing and system delays.
- Watch for eddy current or vibration issues that broaden echoes.
- Use vendor tools (e.g., Prescan, FASTMAP, AutoShim) to optimize shimming.



REAL-WORLD ANALOGY

Think of a group of runners starting together. Uneven terrain makes some go faster than others (dephasing). At halfway, a whistle makes everyone turn around (180° pulse). They run back and meet at the starting line again at time TE (echo).



KEY TAKEAWAY

- 90° tips M into transverse plane → spins dephase.
- 180° pulse reverses phase dispersion.
- Spins rephase at TE → spin echo (max signal).
- Echo amplitude follows T₂ decay (not T₂*).
- Longer TE → stronger T₂ weighting → lower signal.

QUICK REFERENCE

180° Pulse Time	TE/2
Echo Formation	TE
Dominant Relaxation	T ₂
Dephasing Reversed	Yes
T ₂ * Effects Removed	Yes (static)
Diffusion / Flow Effects	No